RESIDUES OF A TROPICAL ZETA FUNCTION FOR CONVEX DOMAINS

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Abstract. For a compact convex domain $\Omega \subset \mathbb{R}^n$, we define an $SL(n,\mathbb{Z})$-invariant zeta function

$$Z_\Omega(s) := \int_\Omega \rho_\Omega(x)^{\,s-n}\, dV,$$

where ρ_Ω is the distance-like function to the boundary determined by primitive lattice supporting lines.

In dimension 2, we introduce two further incarnations of this zeta function. First, we show that the integral zeta function can be rewritten, up to an explicit holomorphic correction term, as a tropical boundary zeta series indexed by Farey neighbors. Second, we show that this boundary series coincides with the zeta function obtained by summing the s-th powers of the coefficients of the Legendre dual in Hata's Schauder basis.

For convex planar domains with sufficiently smooth boundary and everywhere non-zero curvature, we prove that $Z_\Omega(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, with a simple pole there. Its residue is given by

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{3^{5/2}}{2^{5/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \cdot \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

In particular, the first singularity of the tropical zeta function is governed by the equiaffine geometry of the boundary.

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1. The tropical zeta function of a convex domain in the integral form

In this section we define the zeta function of a convex body by an integral over the body itself. In dimension 2, this integral presentation is equivalent, up to an explicit holomorphic correction term, to a boundary Dirichlet series built from primitive lattice directions and Farey neighbors.

1.1. **The tropical distance function.** Let $\Omega \subset \mathbb{R}^n$ be a compact convex body with nonempty interior. For

$$u \in \mathbb{R}^n$$

let

$$h_\Omega(u) := \min_{x\in\Omega} \langle u, x\rangle$$

be the lower support function of Ω. Denote by

$$\mathbb{Z}^n_{\text{prim}} := \{u \in \mathbb{Z}^n \setminus \{0\} : \gcd(u_1, \dots, u_n) = 1\}$$

the set of primitive lattice directions.

Since

$$\langle mu, x\rangle - h_\Omega(mu) = m\big(\langle u, x\rangle - h_\Omega(u)\big) \qquad (m \in \mathbb{N}),$$

it is enough to minimize over primitive lattice vectors.

Definition 1. *For $x \in \Omega^\circ$, define the* tropical distance function *of x by*

$$\rho_\Omega(x) := \min_{u\in\mathbb{Z}^n_{\text{prim}}} \big(\langle u, x\rangle - h_\Omega(u)\big).$$

Indeed, if $x \in \Omega^\circ$, then

$$\langle u, x\rangle - h_\Omega(u) \geq c_x\|u\|$$

for some $c_x > 0$, so the minimum is attained.

The function ρ_Ω is continuous on Ω°, positive on the interior, and can be continuously extended to zero on $\partial\Omega$. It may be viewed as an $SL(n, \mathbb{Z})$-invariant analogue of a distance-to-the-boundary function, built from primitive lattice supporting hyperplanes rather than Euclidean normals.

Definition 2. *The* tropical zeta function (in the integral form) *of Ω is*

$$Z_\Omega(s) := \int_\Omega \rho_\Omega(x)^{\,s-n}\, dx,$$

initially defined for $\Re(s)$ sufficiently large.

write that it is sl 2 z invariant

The normalization $s - n$ is chosen so that $Z_\Omega(n) = \mathrm{vol}(\Omega)$.

we ma have a discussion in appendix about similar zeta functions for convex domains and their zeroes and residues

1.2. **Mellin form.** Let

$$M_\Omega := \max_{x\in\Omega} \rho_\Omega(x)$$

and, for $t \in [0, M_\Omega]$, define

$$\Omega_t := \{x \in \Omega : \rho_\Omega(x) \geq t\}.$$

Then for $\Re(s) > n$, the layer-cake formula gives

$$Z_\Omega(s) = (s-n)\int_0^{M_\Omega} t^{\,s-n-1}\,\mathrm{vol}(\Omega_t)\,dt.$$

Thus $Z_\Omega(s)$ is the Mellin transform of the volume profile of the family $\{\Omega_t\}_{t\geq 0}$.

For every $t > 0$, the set Ω_t is a convex polytope with finitely many facets, each of whose supporting hyperplanes has a primitive lattice normal.

Definition 3. *Let F be an $(n-1)$-dimensional polytope contained in an affine hyperplane*

$$H = \{x \in \mathbb{R}^n : \langle u, x\rangle = m\},$$

where $u \in \mathbb{Z}^n$ is primitive and $m \in \mathbb{Z}$. Its lattice volume *is defined by*

$$\mathrm{vol}_{\mathbb{Z}}(F) := \frac{\mathrm{vol}_{n-1}(F)}{\mathrm{covol}(H\cap\mathbb{Z}^n)},$$

where vol_{n-1} denotes Euclidean $(n-1)$-dimensional volume in H.

Definition 4. *For $t > 0$, the* lattice surface volume *of Ω_t is the sum of the lattice volumes of all facets of Ω_t. We denote it by $P_\Omega(t)$.*

Since $\mathrm{covol}(H\cap\mathbb{Z}^n)$ is equal to the length of the primitive integer normal vector to H, for $t > 0$, one has

$$P_\Omega(t) = -\frac{d}{dt}\,\mathrm{vol}(\Omega_t).$$

shall we prove it? it woluld be better to do so, in appendinx, along with a proof that of omega is a rational polytope then it it has the first pole and the rsidue is proportional to the lattice surface volume of it

Proposition 1 (Lattice surface volume form of the Mellin transform)**.** *For $\Re(s) > n$,*

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-n} P_\Omega(t)\,dt.$$

Proof. Starting from

$$Z_\Omega(s) = (s-n)\int_0^{M_\Omega} t^{s-n-1}\,\mathrm{vol}(\Omega_t)\,dt,$$

integrate by parts:

$$(s-n)\int_0^{M_\Omega} t^{s-n-1}\,\mathrm{vol}(\Omega_t)\,dt = \Big[t^{s-n}\,\mathrm{vol}(\Omega_t)\Big]_0^{M_\Omega} - \int_0^{M_\Omega} t^{s-n}\frac{d}{dt}\,\mathrm{vol}(\Omega_t)\,dt.$$

Since $\mathrm{vol}(\Omega_{M_\Omega}) = 0$ and

$$t^{s-n}\,\mathrm{vol}(\Omega_t) \to 0 \qquad \text{as } t \to 0^+$$

for $\Re(s) > n$, this gives

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-n} \left(-\frac{d}{dt} \operatorname{vol}(\Omega_t) \right) dt.$$

□

In particular, for $n = 2$, $Z_\Omega(s)$ is the Mellin transform of the lattice perimeter profile of the tropical wave fronts Ω_t.

Remark 1. It is natural to ask whether $Z_\Omega(s)$ satisfies a functional equation, or more generally whether its Mellin transform structure reflects a duality on tropical wave fronts. We do not know the answer. survey about zeta functions for convex domains in the appendix?

2. The planar boundary zeta series

We now specialize to the case $n = 2$. Let $\Omega \subset \mathbb{R}^2$ be a convex compact domain. Choose two vertical and two horizontal supporting lines of Ω. They decompose $\partial\Omega$ into four convex arcs. To each such arc Γ, we attach a boundary Dirichlet series.

After a rotation by a multiple of $\pi/2$ and a translation, we may assume that Γ lies in a support triangle bounded by the coordinate axes and a convex curve joining them in the first quadrant. For each primitive vector $(a, b) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0, 0)\}$, let

$$ax + by = \gamma_{a,b}$$

be the supporting line to Ω with inward normal (a, b). If

$$a, b, c, d \in \mathbb{Z}_{\geq 0}, \qquad ad - bc = 1,$$

define

$$f_\Gamma(a, b, c, d) := \left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|.$$

Equivalently, if $\Delta_{a,b,c,d}$ denotes the triangle bounded by the three supporting lines with normals (a, b), (c, d), and $(a + c, b + d)$, then

$$f_\Gamma(a, b, c, d) = \sqrt{2 \operatorname{Area}(\Delta_{a,b,c,d})}.$$

Definition 5. *The* boundary zeta series *attached to* Γ *is*

$$F_\Gamma(s) := \sum_{\substack{a,b,c,d \geq 0 \\ ad-bc=1}} f_\Gamma(a, b, c, d)^s.$$

For the whole boundary, we set

$$F_{\partial\Omega}(s) := \sum_\Gamma F_\Gamma(s),$$

where the sum runs over the four arcs cut out by the chosen horizontal and vertical supporting lines.

The integral zeta function and the boundary Dirichlet series are related by an exact identity.

Theorem 1. *Let $\Omega \subset \mathbb{R}^2$ be a convex compact domain. Assume that, after translation, the four chosen supporting lines are*

$$x = 0, \qquad x = P, \qquad y = 0, \qquad y = Q, \qquad P \geq Q > 0.$$

Then for $\Re(s) > 2$,

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1}.$$

See the proof on page 14.

2.1. **The first singularity and equiaffine perimeter.** We now state the main boundary residue formula.

Theorem 2. *Let Γ be a C^3 convex planar arc with everywhere non-zero curvature. Then $F_\Gamma(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, with a simple pole at $s = \frac{2}{3}$, and*

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}}{4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

See the proof on page 12.

Since $\partial\Omega$ is decomposed into four such arcs, summing Theorem 2 over these four pieces gives

$$\operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s) = \frac{\sqrt{3}}{4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

Combining this with the identity of Theorem 1, which for $\Re(s) > 2$ expresses $Z_\Omega(s)$ as

$$-\frac{F_{\partial\Omega}(s)}{s(s-1)}$$

plus a holomorphic term, we obtain a corresponding residue formula for the interior zeta function $Z_\Omega(s)$. In particular, $Z_\Omega(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$, and

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9}{2}\operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s).$$

Therefore

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9\sqrt{3}}{2\cdot 4^{1/3}\pi^3}\Gamma\left(\frac{1}{3}\right)^3 \mathrm{Length}_{\mathrm{equiaffine}}(\partial\Omega).$$

Thus the first singularity of $Z_\Omega(s)$ is governed by the equiaffine perimeter of $\partial\Omega$. We now recall the basic facts about equiaffine length used in the statement of the theorem.

2.2. **Equiaffine length.** Let $\Gamma : [0,1] \to \mathbb{R}^2$ be a C^2 regular curve,

$$\Gamma(t) = (x(t), y(t)), \qquad \dot\Gamma(t) \neq 0.$$

Its Euclidean curvature is

$$\kappa(t) = \frac{|\det(\dot\Gamma(t), \ddot\Gamma(t))|}{\|\dot\Gamma(t)\|^3},$$

and its equiaffine arclength element is

$$ds_{\text{equiaffine}} = |\det(\dot{\Gamma}(t), \ddot{\Gamma}(t))|^{1/3}\, dt.$$

Accordingly,

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \int_\Gamma \kappa^{1/3}\, ds = \int_0^1 |\det(\dot{\Gamma}(t), \ddot{\Gamma}(t))|^{1/3}\, dt.$$

If Γ is written locally as the graph of a convex C^2-function $y = g(x)$, then

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \int (g''(x))^{1/3}\, dx.$$

There is also a geometric interpretation in terms of support triangles: if Γ is partitioned into sufficiently small arcs and Δ_i denotes the support triangle corresponding to the i-th arc, then

$$\text{Length}_{\text{equiaffine}}(\Gamma) = \lim_{\max_i |t_i - t_{i-1}| \to 0} \sum_i \big(2\ \text{Area}(\Delta_i)\big)^{1/3}. \tag{1}$$

3. Farey intervals and the zeta function

Definition 6 (Farey interval)**.** *A* Farey interval *is an interval*

$$I = [c/d,\ a/b] \subset [0,1]$$

with $b, d \in \mathbb{N}$, $a, c \in \mathbb{Z}$ *satisfying* $0 \le c \le d$, $0 \le a \le b$, $\gcd(a,b) = \gcd(c,d) = 1$, *and*

$$ad - bc = 1.$$

We write its mediant as $m(I) = \frac{a+c}{b+d}$.

Definition 7 (Hata coefficient and its renormalization)**.** *Let* $f \in C^2([0,1])$. *For* $I = [c/d, a/b]$ *define*

$$c_I(f) := f\left(\frac{a+c}{b+d}\right) - \frac{b}{b+d} f\left(\frac{a}{b}\right) - \frac{d}{b+d} f\left(\frac{c}{d}\right). \tag{2}$$

Define

$$T_I(f) := (b+d)\, c_I(f).$$

Definition 8 (Farey zeta function)**.** *For* $s \in \mathbb{C}$ *with* $\Re(s)$ *large, define*

$$Z_f(s) := \sum_I |T_I(f)|^s,$$

where the sum runs over all Farey intervals $I \subset [0,1]$.

Remark 2 (On residues)**.** Throughout, $\text{Res}_{s=2/3}$ denotes the residue of a *meromorphic continuation* of the relevant function to a punctured neighborhood of $s = \frac{2}{3}$. Accordingly, whenever we compute a residue we also prove that, after subtracting the explicit polar part, the remainder is holomorphic on some disk $|s - \frac{2}{3}| < \eta$.

Remark 3. Removing or adding finitely many intervals (e.g. those adjacent to 0/1 or 1/1) changes $Z_f(s)$ by an entire function. Hence it does not affect the residue at $s = \frac{2}{3}$.

Assume from now on that

$$f \in C^3([0,1]) \quad \text{and} \quad 0 < m \le |f''(x)| \le M \text{ for all } x \in [0,1]. \tag{3}$$

Theorem 3 (Residue formula)**.** *Assume* (3)*. Then $Z_f(s)$ admits a meromorphic continuation to a disk $|s-\frac{2}{3}|<\eta$ (for some $\eta>0$) with a simple pole at $s=\frac{2}{3}$. Moreover,*

$$\mathrm{Res}_{s=2/3}\, Z_f(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3}\int_0^1 |f''(v)|^{2/3}\,dv.$$

See the proof on page 3. We first reduce the geometric quantity $T_I(f)$ to the local curvature f'' on the interval I. This leads to an endpoint model $Z_f^{\mathrm{end}}(s)$, which has the same residue at $s=\frac{2}{3}$. We then reparametrize Farey intervals by coprime denominator pairs, reorganize the series by the first denominator b, and show that the inner sums admit an equidistributed asymptotic with a power-saving error. The pole then comes entirely from the classical Dirichlet series $\sum_{b\geq 1}\varphi(b)b^{-3s}$.

To every Farey interval I, Hata associates the piecewise-linear function

$$S_I(x) := \frac{b+d}{2}\Big(|a-bx|+|c-dx|-|a+c-(b+d)x|\Big).$$

Lemma 1. *For every Farey interval $I=\left[\frac{c}{d},\frac{a}{b}\right]$, the function S_I satisfies*

$$S_I(x)\geq 0,\qquad \mathrm{supp}(S_I)=I,\qquad S_I\Big(\frac{c}{d}\Big)=S_I\Big(\frac{a}{b}\Big)=0,\qquad S_I(m_I)=1.$$

Moreover, S_I is piecewise linear and attains its unique maximum at the mediant m_I.

Theorem 4 (Hata)**.** *Every continuous function $f\in C[0,1]$ admits the expansion*

$$f(x)=f(0)+(f(1)-f(0))x+\sum_I c_I(f)\,S_I(x),$$

where the sum is taken over all Farey intervals in Stern–Brocot order and converges uniformly on $[0,1]$. The coefficient of S_I is given by (2)*.*

For our purposes, the importance of Hata's coefficient is that after the normalization $T_I(f)=(b+d)c_I(f)$, it behaves like a second-order local quantity attached to the interval I. Thus the zeta function $Z_f(s)$ is expected to be controlled by the local curvature f'', with a geometric weight depending only on the Farey denominators $b,d,b+d$. The next lemma makes this heuristic exact.

Lemma 2 (Mean-value formula for $T_I(f)$, proof on page A.3)**.** *Let $I=[c/d,a/b]$ be a Farey interval and $f\in C^2([0,1])$. Then there exists $\xi_I\in(c/d,a/b)$ such that*

$$T_I(f)=-\frac{f''(\xi_I)}{2\,bd(b+d)}.$$

Thus $c_I(f)$ is a weighted discrete second derivative of u along the Farey interval I. Lemma 2 shows that each summand of $Z_f(s)$ is governed by the value of f'' at some interior point ξ_I, multiplied by the universal arithmetic weight $(bd(b+d))^{-1}$. To isolate the arithmetic structure, we now replace the interior point ξ_I by one endpoint. This produces a simpler model series $Z_f^{\mathrm{end}}(s)$, and the next lemma shows that this modification does not change the residue at $s=\frac{2}{3}$.

Define

$$Z_f^{\mathrm{end}}(s) := 2^{-s} \sum_{I=[c/d,a/b]} \frac{|f''(a/b)|^s}{(bd(b+d))^s}.$$

Replacing ξ_I by an endpoint does not change the residue, as it is proved in the next lemma.

Lemma 3 (Proof on page 19)**.** *There exists $\eta_0 > 0$ such that $Z_f(s) - Z_f^{\mathrm{end}}(s)$ is holomorphic on the disk $|s-\frac{2}{3}| < \eta_0$. In particular,* $\mathrm{Res}_{s=2/3} Z_f = \mathrm{Res}_{s=2/3} Z_f^{\mathrm{end}}$.

Lemma 4 (Parametrization by modular inverses)**.** *Farey intervals $I = [c/d, a/b]$ (with $b, d \geq 1$) are in bijection with coprime pairs $(b, d) \in \mathbb{N}^2$ via*

$$a \equiv d^{-1} \pmod{b}, \quad 1 \leq a \leq b, \qquad c = \frac{ad-1}{b}.$$

Proof. If $ad - bc = 1$, then $ad \equiv 1 \pmod{b}$, so $(b, d) = 1$ and $a \equiv d^{-1} \pmod{b}$. Conversely, for coprime (b, d) choose a with $ad \equiv 1 \pmod{b}$ and define $c = (ad-1)/b \in \mathbb{Z}$; then $ad - bc = 1$. □

The parametrization by (b, d) separates the Farey combinatorics into an outer denominator b and an inner progression $d \equiv \overline{r} \pmod{b}$. After writing $d = kb + r$, the dependence on k becomes one-dimensional and is absorbed into a universal kernel $H_s(r/b)$. In this way the original Farey sum is reorganized into outer sums over b and inner reduced-residue sums over r.

For $\Re(s) > \frac{1}{2}$ and $u \in (0, 1]$, define

$$H_s(u) := \sum_{k=0}^{\infty} (k+u)^{-s}(k+1+u)^{-s}.$$

For $\Re(s) > \frac{1}{2}$, using Lemma 4 and writing $d = kb + r$ with $(r, b) = 1$, one obtains

$$Z_f^{\mathrm{end}}(s) = 2^{-s} \sum_{b \geq 1} b^{-3s} \sum_{\substack{1 \leq r \leq b \\ (r,b)=1}} |f''(\overline{r}/b)|^s \, H_s(r/b), \tag{4}$$

where $\overline{r}$ denotes the inverse of r modulo b in $\{1, \ldots, b\}$.

Lemma 5 (Integral of H_s)**.** *For $\frac{1}{2} < \Re(s) < 1$,*

$$\int_0^1 H_s(u)\, du = \int_0^{\infty} t^{-s}(1+t)^{-s}\, dt = \frac{\Gamma(1-s)\Gamma(2s-1)}{\Gamma(s)}.$$

In particular, $\int_0^1 H_{2/3}(u)\, du = \Gamma(1/3)^2/\Gamma(2/3)$.

Proof. For $\Re(s) > \frac{1}{2}$, absolute convergence gives

$$\int_0^1 H_s(u)\, du = \sum_{k \geq 0} \int_0^1 (k+u)^{-s}(k+1+u)^{-s}\, du$$

by Fubini's theorem. Substitute $t = k + u$ in each term and note that the intervals $[k, k+1]$ partition $[0, \infty)$. This yields $\int_0^{\infty} t^{-s}(1+t)^{-s} dt$, which equals the Beta integral. □

The decomposition (4) reduces the residue problem to understanding, for each fixed b, the reduced-residue average of the product $H_s(r/b)\,|f''(\overline{r}/b)|^s$. We denote this inner sum by $\Sigma_b(s)$. The key analytic step is that $\Sigma_b(s)$ admits an asymptotic formula with a power-saving error, uniformly in s on compact subsets of $\{1/2 < \Re(s) < 1\}$.

Define

$$\Sigma_b(s) := \sum_{\substack{1\le r\le b\\ (r,b)=1}} H_s(r/b)\,|f''(\overline{r}/b)|^s.$$

We prove the following equidistribution estimate for $\Sigma_b(s)$ with a power saving.

Proposition 2 (Proof on page 25)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$ and assume* (3)*. Then for every $\varepsilon > 0$ and $s \in K$ we have*

$$\Sigma_b(s) = \varphi(b)\left(\int_0^1 H_s(u)\,du\right)\left(\int_0^1 |f''(v)|^s\,dv\right) + O_{f,K,\varepsilon}\big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\big),$$

with implied constant uniform for $s \in K$.

At this point the analytic difficulty is essentially finished. Indeed, Proposition 2 turns $Z_f^{\mathrm{end}}(s)$ into an explicit main term involving $\sum_{b\ge1}\varphi(b)b^{-3s}$, plus an error which is holomorphic near $s = \frac{2}{3}$. Thus the pole and its residue come from the Euler-factor Dirichlet series for φ.

Proof of Theorem 3. By Lemma 3, it suffices to compute the residue of $Z_f^{\mathrm{end}}(s)$. Fix a disk $D := \{s : |s - 2/3| \le \eta\}$ with $0 < \eta < 1/30$ and set $\sigma_- := 2/3 - \eta$, $\sigma_+ := 2/3 + \eta$.

From (4) and Proposition 2 (applied on a compact set containing D but lying in $\{1/2 < \Re(s) < 1\}$), we get for $s \in D$:

$$Z_f^{\mathrm{end}}(s) = 2^{-s}\Big(\int_0^1 H_s\Big)\Big(\int_0^1 |f''(v)|^s\Big)\sum_{b\ge1}\frac{\varphi(b)}{b^{3s}} + E(s),$$

where

$$E(s) = 2^{-s}\sum_{b\ge1} b^{-3s}\,R_b(s), \qquad |R_b(s)| \ll_{f,D,\varepsilon} b^{\frac{1+\Re(s)}{2}+\varepsilon}.$$

In particular, for $s \in D$,

$$|b^{-3s}R_b(s)| \ll_{f,D,\varepsilon} b^{-3\sigma_- + \frac{1+\sigma_+}{2}+\varepsilon} = b^{-\frac{7}{6}+\frac{7\eta}{2}+\varepsilon}.$$

Choose $\varepsilon > 0$ so small that $-\frac{7}{6} + \frac{7\eta}{2} + \varepsilon < -1$ (possible since $\eta < 1/30$). Then $\sum_{b\ge1} b^{-7/6+7\eta/2+\varepsilon}$ converges. Hence the series defining $E(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, $E(s)$ is holomorphic on the interior of D.

The factor $A(s) := 2^{-s}\big(\int_0^1 H_s\big)\big(\int_0^1 |f''(v)|^s\big)$ is holomorphic on $\{1/2 < \Re(s) < 1\}$ by Lemma 5 and dominated convergence. The Dirichlet series identity $\sum_{n\ge1}\varphi(n)n^{-w} = \zeta(w-1)/\zeta(w)$ valid for $\Re(w) > 2$ gives a meromorphic continuation of $\sum_{b\ge1}\varphi(b)b^{-3s}$ near $s = 2/3$, with a simple pole at

$3s = 2$ and residue $1/(3\zeta(2))$. Therefore $Z_f^{\mathrm{end}}(s)$ (hence $Z_f(s)$) is meromorphic on a neighborhood of $2/3$ with

$$\mathrm{Res}_{s=2/3}\, Z_f(s) = A(2/3) \cdot \frac{1}{3\zeta(2)}.$$

Now $A(2/3) = 2^{-2/3}\left(\int_0^1 H_{2/3}\right)\left(\int_0^1 |f''(v)|^{2/3} dv\right)$. By Lemma 5, $\int_0^1 H_{2/3} = \Gamma(1/3)^2/\Gamma(2/3)$. Using $\zeta(2) = \pi^2/6$ and $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$, the constant simplifies to $\sqrt{3}\,\Gamma(1/3)^3/(2^{2/3}\pi^3)$. □

3.1. **Hata's basis and Legendre duality.** We now explain that the boundary zeta series is, up to finitely many harmless terms, exactly the zeta series attached to the Hata coefficients of the Legendre dual. Thus Theorem 2 is a direct consequence of Theorem 3 applied to a suitable dual function.

Let $\Gamma \subset \partial\Omega$ be one of the four arcs cut out by the chosen horizontal and vertical supporting lines. After an $SL(2,\mathbb{Z})$ transformation followed by a translation, we may assume that Γ is the graph of a convex decreasing C^3-function

$$y = g(x), \qquad x \in [0,1],$$

and that the parameter

$$u = -g'(x)$$

runs through $[0,1]$ along Γ.

Define

$$\widetilde{g}(u) := g^*(-u), \qquad u \in [0,1],$$

where

$$g^*(\alpha) := \sup_{x\in\mathbb{R}}(\alpha x - g(x))$$

is the Legendre transform of g.

A supporting line to $y = g(x)$ with primitive integer normal $(a,b) \in \mathbb{Z}^2_{\geq 0} \setminus \{(0,0)\}$ has equation

$$ax + by = \gamma_{a,b}.$$

If $b > 0$, then

$$y = -\frac{a}{b}x + \frac{\gamma_{a,b}}{b}.$$

Its slope is $-a/b$, and comparison with the Legendre representation

$$y = \alpha x - g^*(\alpha)$$

shows that

$$\widetilde{g}\Big(\frac{a}{b}\Big) = g^*\Big(-\frac{a}{b}\Big) = -\frac{\gamma_{a,b}}{b}.$$

Thus the values of $\widetilde{g}$ at rational points encode the support data of Ω.

Lemma 6 (Identification of the geometric weights)**.** *Let*

$$I = \left[\frac{c}{d}, \frac{a}{b}\right]$$

be a Farey interval, and let $v_I = (a+c, b+d)$ be the associated primitive vector. Then

$$T_I(\widetilde{g}) := (b+d)c_I(\widetilde{g}) = -\gamma_{a+c,b+d} + \gamma_{a,b} + \gamma_{c,d}.$$

In particular,

$$|T_I(\widetilde{g})| = f_\Gamma(a,b,c,d).$$

Proof. By the coefficient formula,

$$\begin{aligned} c_I(\widetilde{g}) &= \widetilde{g}\Big(\frac{a+c}{b+d}\Big) - \frac{b}{b+d}\widetilde{g}\Big(\frac{a}{b}\Big) - \frac{d}{b+d}\widetilde{g}\Big(\frac{c}{d}\Big) \\ &= -\frac{\gamma_{a+c,b+d}}{b+d} + \frac{\gamma_{a,b}}{b+d} + \frac{\gamma_{c,d}}{b+d}, \end{aligned}$$

and multiplying by $b+d$ gives the formula. □

The previous lemma shows that the geometric Dirichlet series attached to Γ is exactly the Hata-coefficient zeta series for the dual function $\widetilde{g}$, up to finitely many exceptional terms near the ends of the arc. As usual, such finitely many terms contribute an entire function and therefore do not affect residues.

Corollary 1. *There exists an entire function $E_\Gamma(s)$ such that*

$$F_\Gamma(s) = Z_{\widetilde{g}}(s) + E_\Gamma(s).$$

In particular,

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \operatorname*{Res}_{s=2/3} Z_{\widetilde{g}}(s).$$

It remains to identify the integral appearing in Theorem 3 with the equiaffine length of Γ.

Lemma 7 (Legendre duality and equiaffine length)**.** *The function $\widetilde{g}$ satisfies the assumptions of Theorem 3. Moreover,*

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\,du = \int_0^1 (g''(x))^{1/3}\,dx = \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

Proof. Since $g \in C^3$ is convex and has a non-zero curvature on a compact interval, there exist constants $0 < m \le M < \infty$ such that

$$0 < m \le g''(x) \le M \qquad (x \in [0,1]).$$

Standard Legendre duality implies that $\widetilde{g} \in C^3([0,1])$ and, if $u = -g'(x)$, then

$$\widetilde{g}''(u) = \frac{1}{g''(x)}.$$

Also,

$$du = -g''(x)\,dx.$$

Since $u = -g'(x)$ decreases from 1 to 0 as x increases from 0 to 1, we have

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\,du = \int_1^0 (g''(x))^{-2/3}\,(-g''(x))\,dx = \int_0^1 (g''(x))^{1/3}\,dx.$$

This proves the claim.

□

Proof of Theorem 2. By Corollary 1, the residue of $F_\Gamma(s)$ at $s = \frac{2}{3}$ is the same as the residue of $Z_{\widetilde{g}}(s)$. Applying Theorem 3 with $f = \widetilde{g}$, we obtain

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3} \int_0^1 (\widetilde{g}''(u))^{2/3}\, du.$$

By Lemma 7,

$$\int_0^1 (\widetilde{g}''(u))^{2/3}\, du = \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

Hence

$$\operatorname*{Res}_{s=2/3} F_\Gamma(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3}\, \mathrm{Length}_{\mathrm{equiaffine}}(\Gamma).$$

This proves the theorem. □

3.2. **Geometric meaning of the terms of the boundary zeta series.** The summands of the boundary zeta series admit a simple geometric interpretation in terms of small triangles cut out by supporting lines.

Let (a, b) and (c, d) be a unimodular pair, so that $ad - bc = 1$, and let $\Delta_{a,b,c,d}$ be the triangle bounded by the three supporting lines with primitive normals

$$(a, b), \qquad (c, d), \qquad (a + c, b + d).$$

Then

$$\mathrm{Area}(\Delta_{a,b,c,d}) = \frac{1}{2}\Big(\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\Big)^2.$$

Indeed, after applying an $SL(2, \mathbb{Z})$ transformation and then a translation, we may reduce to the case

$$(a, b) = (1, 0), \qquad (c, d) = (0, 1), \qquad \gamma_{a,b} = \gamma_{c,d} = 0.$$

In these coordinates, the three supporting lines are

$$x = 0, \qquad y = 0, \qquad x + y = \gamma_{a+c,b+d},$$

so $\Delta_{a,b,c,d}$ is a right isosceles triangle whose legs have length

$$\big|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\big|.$$

Its area is therefore exactly

$$\frac{1}{2}\Big(\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\Big)^2.$$

Consequently, the terms in

$$F_\Gamma(s) = \sum_{\substack{a,b,c,d \geq 0 \\ ad-bc=1}} f_\Gamma(a, b, c, d)^s$$

are given by

$$f_\Gamma(a, b, c, d) = \big|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\big| = \sqrt{2\ \mathrm{Area}(\Delta_{a,b,c,d})}.$$

Thus the coefficients of Hata's expansion of the Legendre dual have a direct geometric meaning: they measure the sizes of the support triangles determined by neighboring primitive normals.

Theorem 2 shows that the first singularity of the boundary Dirichlet series is governed by the asymptotic distribution of these small support triangles. Since

$$f_\Gamma(a,b,c,d) = \sqrt{2\ \mathrm{Area}(\Delta_{a,b,c,d})},$$

the residue at $s = \frac{2}{3}$ may be viewed as encoding a 1/3-power summation law for the areas of these triangles. This is precisely the exponent that appears in the formula (1) for the equiaffine length.

Appendix A. Basic geometry of two-dimensional case

A.1. **A local model.** We begin with the simplest support configuration, namely the rectangular box determined by the horizontal and vertical supporting lines. The idea is that the global integral identity will be obtained by starting from this box and then inserting the remaining supporting lines one by one. Each new supporting line cuts off a single triangular region, and the next lemma computes the contribution of the initial rectangle.

Let

$$R = [0,P] \times [0,Q], \qquad P \geq Q > 0,$$

and define

$$\rho_R(x,y) := \min(x, P-x, y, Q-y).$$

Lemma 8 (Rectangle identity)**.** *For* $\Re(s) > 2$,

$$s(s-1)\int_R \rho_R(x,y)^{s-2}\,dx\,dy = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1}.$$

Proof. For $0 \leq t \leq Q/2$, the superlevel set of ρ_R is

$$\{\rho_R \geq t\} = [t, P-t] \times [t, Q-t],$$

and therefore

$$\mathrm{Area}\{\rho_R \geq t\} = (P-2t)(Q-2t).$$

By the layer-cake formula,

$$\int_R \rho_R(x,y)^{s-2}\,dx\,dy = (s-2)\int_0^{Q/2} t^{s-3}(P-2t)(Q-2t)\,dt.$$

Expanding the integrand gives

$$(s-2)\int_0^{Q/2} t^{s-3}\big(PQ - 2(P+Q)t + 4t^2\big)\,dt.$$

Integrating term by term and simplifying yields

$$s(s-1)\int_R \rho_R(x,y)^{s-2}\,dx\,dy = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1},$$

as claimed. □

We next isolate the local effect of inserting a single additional supporting line. After a unimodular affine normalization, every such local move is reduced to cutting the first quadrant by a line of the form $x + y = \lambda$. The following identity computes exactly the change of the integral under this elementary operation.

Lemma 9 (one-cut identity)**.** *Let $\lambda > 0$, and define*

$$\phi_-(x,y) := \min(x,y) \qquad \text{on } \{x \geq 0,\ y \geq 0\},$$

and

$$\phi_+(x,y) := \min(x,y,x+y-\lambda) \qquad \text{on } \{x \geq 0,\ y \geq 0,\ x+y \geq \lambda\}.$$

Then for $\Re(s) > 2$,

$$\int_{\{x+y<\lambda\}} \phi_-(x,y)^{\,s-2}\,dx\,dy + \int_{\{x\geq 0,\ y\geq 0,\ x+y\geq\lambda\}} \Big(\phi_-(x,y)^{\,s-2} - \phi_+(x,y)^{\,s-2}\Big)\,dx\,dy = \frac{\lambda^s}{s(s-1)}.$$

Proof. For $t \geq 0$, the difference of the superlevel sets appearing on the left-hand side is

$$E_t := \{x \geq t,\ y \geq t,\ x+y \leq \lambda + t\}.$$

After the change of variables $x' = x - t$, $y' = y - t$, this becomes

$$\{x' \geq 0,\ y' \geq 0,\ x' + y' \leq \lambda - t\}.$$

Hence, for $0 \leq t \leq \lambda$,

$$\mathrm{Area}(E_t) = \frac{(\lambda - t)^2}{2},$$

while for $t > \lambda$ the set E_t is empty.

Applying the layer-cake formula, we find that the left-hand side equals

$$(s-2)\int_0^\lambda t^{s-3}\frac{(\lambda-t)^2}{2}\,dt.$$

Substituting $t = \lambda u$ gives

$$\frac{s-2}{2}\lambda^s \int_0^1 u^{s-3}(1-u)^2\,du.$$

Since

$$\int_0^1 u^{s-3}(1-u)^2\,du = B(s-2,3) = \frac{2}{s(s-1)(s-2)},$$

we obtain

$$\frac{\lambda^s}{s(s-1)}.$$

□

A.2. **The integral–boundary identity.** We now assemble the global identity. Starting from the rectangular support box R, we insert the remaining supporting lines in Stern–Brocot order. The rectangle identity gives the initial contribution, while the one-cut identity shows that each newly inserted support line subtracts exactly one term of the boundary Dirichlet series.

Proof of theorem 1. Let $R = [0,P] \times [0,Q]$. By Lemma 8,

$$s(s-1)\int_R \rho_R(x,y)^{\,s-2}\,dx\,dy = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1}.$$

Now construct a sequence of convex polygons $\Omega^{(N)}$ by inserting supporting lines in Stern–Brocot order along each of the four arcs of $\partial\Omega$, starting from the rectangle R. At each step, one inserts the mediant supporting line

between two neighboring supporting lines with inward primitive normals (a, b) and (c, d), where $ad - bc = 1$.

After an affine change of coordinates in $SL(2, \mathbb{Z})$, the two old supporting lines become

$$x = 0, \qquad y = 0,$$

and the new mediant supporting line becomes

$$x + y = \lambda, \qquad \lambda = \left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|.$$

Because the change of coordinates lies in $SL(2, \mathbb{Z})$, it preserves the Euclidean area form. Thus the local change of the integral is exactly the one computed in Lemma 9. It follows that each insertion decreases the current value of the integral by

$$\frac{1}{s(s-1)}\left|\gamma_{a,b} + \gamma_{c,d} - \gamma_{a+c,b+d}\right|^s.$$

Hence, after finitely many insertions, we obtain

$$s(s-1)Z_{\Omega^{(N)}}(s) = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1} - \sum_{\Delta \in \mathcal{T}_N} \big(\sqrt{2\ \mathrm{Area}(\Delta)}\big)^s,$$

where $\mathcal{T}_N$ is the finite set of support triangles created up to level N.

As $N \to \infty$, the polygons $\Omega^{(N)}$ decrease to Ω, and the corresponding distance functions $\rho_{\Omega^{(N)}}$ decrease pointwise to ρ_Ω. Since $\Re(s) > 2$, the functions $\rho_{\Omega^{(N)}}^{s-2}$ are dominated by $\rho_R^{\Re(s)-2} \in L^1(R)$. Therefore, by dominated convergence,

$$\int_{\Omega^{(N)}} \rho_{\Omega^{(N)}}(x,y)^{\,s-2}\,dx\,dy \longrightarrow \int_\Omega \rho_\Omega(x,y)^{\,s-2}\,dx\,dy.$$

On the other hand, for $\Re(s) > 2$, the finite sums over $\mathcal{T}_N$ converge absolutely to $F_{\partial\Omega}(s)$. Passing to the limit yields

$$s(s-1)Z_\Omega(s) = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1} - F_{\partial\Omega}(s).$$

□

In particular, there exists a holomorphic function $H_\Omega(s)$, namely

$$H_\Omega(s) = 8\Big(\frac{Q}{2}\Big)^s + 2s(P-Q)\Big(\frac{Q}{2}\Big)^{s-1},$$

such that

$$s(s-1)Z_\Omega(s) = -F_{\partial\Omega}(s) + H_\Omega(s).$$

Thus, in dimension 2, the interior zeta function is determined by the boundary Dirichlet series up to an explicit holomorphic correction term coming from the initial rectangular box. In particular, all nontrivial singular behavior of $Z_\Omega(s)$ is already encoded on the boundary.

A.3. **Exact mean-value identity.**

Lemma 10 (Two-point second-order mean value formula)**.** *Let $f \in C^2([0,1])$, let $x \neq y$ in $[0,1]$, and let $\lambda \in (0,1)$. Put*

$$m = \lambda x + (1-\lambda)y.$$

Then there exists ξ between x and y such that

$$f(m) - \lambda f(x) - (1-\lambda)f(y) = -\frac{1}{2}\lambda(1-\lambda)f''(\xi)(x-y)^2.$$

Proof. Define

$$\varphi(t) := f\big(y + t(x-y)\big), \qquad t \in [0,1].$$

Then $\varphi \in C^2([0,1])$, and

$$\varphi(\lambda) = f(m), \qquad \varphi(1) = f(x), \qquad \varphi(0) = f(y).$$

Thus it is enough to prove that there exists $\eta \in (0,1)$ such that

$$\varphi(\lambda) - \lambda\varphi(1) - (1-\lambda)\varphi(0) = -\frac{1}{2}\lambda(1-\lambda)\varphi''(\eta).$$

Set

$$L(t) := \varphi(0) + t\big(\varphi(1) - \varphi(0)\big),$$

and let

$$A := \varphi(\lambda) - L(\lambda).$$

Consider the auxiliary function

$$\psi(t) := \varphi(t) - L(t) - A\,\frac{t(t-1)}{\lambda(\lambda-1)}.$$

Then

$$\psi(0) = \psi(\lambda) = \psi(1) = 0.$$

By Rolle's theorem, there exist $u_1 \in (0,\lambda)$ and $u_2 \in (\lambda,1)$ such that

$$\psi'(u_1) = \psi'(u_2) = 0.$$

Applying Rolle's theorem once more to ψ', we obtain $\eta \in (u_1,u_2) \subset (0,1)$ such that

$$\psi''(\eta) = 0.$$

Since $L'' = 0$ and $\big(t(t-1)\big)'' = 2$, this gives

$$0 = \psi''(\eta) = \varphi''(\eta) - A\,\frac{2}{\lambda(\lambda-1)}.$$

Hence

$$A = \frac{1}{2}\,\lambda(\lambda-1)\varphi''(\eta) = -\frac{1}{2}\,\lambda(1-\lambda)\varphi''(\eta).$$

Finally,

$$\varphi''(t) = f''\big(y + t(x-y)\big)(x-y)^2.$$

Therefore, with

$$\xi := y + \eta(x-y),$$

we get

$$f(m) - \lambda f(x) - (1-\lambda)f(y) = -\frac{1}{2}\lambda(1-\lambda)f''(\xi)(x-y)^2.$$

□

Proof of Lemma 2. Apply Lemma 10 with

$$x = \frac{a}{b}, \qquad y = \frac{c}{d}, \qquad \lambda = \frac{b}{b+d}.$$

Since $ad - bc = 1$, we have

$$x - y = \frac{ad - bc}{bd} = \frac{1}{bd} \neq 0,$$

so the lemma applies. Moreover,

$$m = \lambda x + (1-\lambda)y = \frac{b}{b+d}\frac{a}{b} + \frac{d}{b+d}\frac{c}{d} = \frac{a+c}{b+d} = m(I).$$

Hence

$$c_I(f) = f(m) - \frac{b}{b+d} f\Big(\frac{a}{b}\Big) - \frac{d}{b+d} f\Big(\frac{c}{d}\Big) = -\frac{1}{2}\lambda(1-\lambda) f''(\xi_I)(x-y)^2$$

for some $\xi_I \in (c/d, a/b)$. Multiplying by $b+d$, we obtain

$$(b+d)c_I(f) = -\frac{1}{2}(b+d)\lambda(1-\lambda) f''(\xi_I)(x-y)^2.$$

Now

$$\lambda(1-\lambda) = \frac{bd}{(b+d)^2}, \qquad (x-y)^2 = \frac{(ad-bc)^2}{b^2d^2} = \frac{1}{b^2d^2}.$$

Substituting these identities gives

$$(b+d)c_I(f) = -\frac{f''(\xi_I)}{2\,bd(b+d)},$$

as claimed. □

Appendix B. Asymptotics of the lattice perimeter of tropical wave fronts

The Mellin interpretation is given by Proposition 1: the function $Z_\Omega(s)$ is the Mellin transform of the lattice perimeter profile of the tropical wave fronts Ω_t. More precisely, if

$$P_\Omega(t) := -\frac{d}{dt}\,\mathrm{Area}(\Omega_t)$$

exists and satisfies

$$P_\Omega(t) = C_\Omega\, t^{1/3} + O(t^{1/3+\eta}) \qquad (t \to 0^+)$$

for some $\eta > 0$, then Proposition 1 gives

$$Z_\Omega(s) = \int_0^{M_\Omega} t^{s-2} P_\Omega(t)\,dt = \frac{C_\Omega}{s - \frac{2}{3}} + \text{holomorphic terms}.$$

In particular,

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = C_\Omega.$$

Equivalently,

$$\mathrm{Area}(\Omega) - \mathrm{Area}(\Omega_t) = \frac{3}{4}\,C_\Omega\, t^{4/3} + O(t^{4/3+\eta}).$$

Hence the asymptotic law

$$P_\Omega(t) \sim C_\Omega\, t^{1/3}$$

is the geometric meaning of the first singularity of $Z_\Omega(s)$, and the coefficient C_Ω is proportional to the equiaffine perimeter of $\partial\Omega$.

Recall that for $t \geq 0$ we denote by

$$\Omega_t := \{x \in \Omega : \rho_\Omega(x) \geq t\}$$

the tropical wave front of Ω at time t. In this section we study the asymptotic behavior of its lattice perimeter as $t \to 0^+$, under the same smoothness assumptions on $\partial\Omega$ as in the residue theorem at $s = \frac{2}{3}$.

Let $\widehat{\Omega}$ be the minimal model of Ω. For $t > 0$, let $\widehat{\Omega}^t$ be the polygon obtained from $\widehat{\Omega}$ by removing all corners of size greater than t. In particular,

$$\widehat{\Omega}^t = \widehat{\Omega} \qquad \text{for } t \text{ sufficiently large.}$$

Lemma 11. *For $0 \leq u < t$, the dual fan of $(\widehat{\Omega}^t)_u$ is constant.*

Since $\widehat{\Omega}^t$ is a unimodular polygon, it defines a symplectic toric surface. Let K_t be its canonical class and ω_t the corresponding symplectic form. The canonical evolution equation and the lemma above imply that the class of the symplectic form corresponding to $(\widehat{\Omega}^t)_u$ for $0 \leq u < t$ is

$$[\omega_t] + uK_t.$$

Hence the lattice perimeter of $(\widehat{\Omega}^t)_u$ is

$$\mathrm{Per}_{\mathbb{Z}}\big((\widehat{\Omega}^t)_u\big) = \mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) - u\,K_t^2.$$

We also need the compatibility relation between the truncated models and the actual wave fronts.

Lemma 12. *For every $u > 0$,*

$$(\widehat{\Omega}^u)_u = \Omega_u.$$

Now let

$$A_\Omega(t) := \#\{\Delta : \ \mathrm{size}(\Delta) \geq t\},$$

where the count runs over the support triangles arising in the carving process. Then $\widehat{\Omega}^t$ is obtained from $\widehat{\Omega}$ by performing $A_\Omega(t)$ toric blow-ups, and each blow-up decreases K^2 by 1. Therefore

$$K_t^2 = K_{\widehat{\Omega}}^2 - A_\Omega(t).$$

Let

$$r_\Omega := \operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s).$$

Since $F_{\partial\Omega}(s)$ is the Dirichlet generating series of the support-triangle sizes, Perron's formula gives

$$A_\Omega(t) = \frac{3}{2}\, r_\Omega\, t^{-2/3} + o(t^{-2/3}) \qquad (t \to 0^+).$$

On the other hand, the lattice perimeter of Ω itself is zero, since $\partial\Omega$ has no straight edges. Thus the lattice perimeter of $\widehat{\Omega}^t$ is the tail of the sum of triangle sizes:

$$\mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^t) = -\int_0^t u\, dA_\Omega(u).$$

Integrating by parts, we obtain

$$\mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^{\,t}) = -tA_\Omega(t) + \int_0^t A_\Omega(u)\,du = 3r_\Omega\, t^{1/3} + o(t^{1/3}).$$

Using the toric evolution formula at $u = t$, we get

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \mathrm{Per}_{\mathbb{Z}}(\widehat{\Omega}^{\,t}) + tA_\Omega(t) + O(t),$$

and hence

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \frac{9}{2}\, r_\Omega\, t^{1/3} + o(t^{1/3}).$$

Since

$$\operatorname*{Res}_{s=2/3} Z_\Omega(s) = \frac{9}{2} \operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s) = \frac{9}{2}\, r_\Omega,$$

this may be rewritten as follows.

Theorem 5. *Let $\Omega \subset \mathbb{R}^2$ be a compact convex tropical domain with sufficiently smooth boundary and with non-zero curvature. Then, as $t \to 0^+$,*

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \operatorname*{Res}_{s=2/3} Z_\Omega(s)\, t^{1/3} + o(t^{1/3}).$$

Equivalently,

$$\mathrm{Per}_{\mathbb{Z}}(\Omega_t) = \frac{9}{2} \operatorname*{Res}_{s=2/3} F_{\partial\Omega}(s)\, t^{1/3} + o(t^{1/3}).$$

Since the residue is proportional to the equiaffine perimeter of $\partial\Omega$, the leading asymptotic of the lattice perimeter of the tropical wave fronts recovers the equiaffine perimeter of Ω.

Appendix C. Analytic proof of Proposition 2

C.0.1. *Regularity hypotheses and analytic dependence on s.* For $s \in \mathbb{C}$ define the 1-periodic function

$$B_s(x) := |f''(x)|^s = \exp\bigl(s \log |f''(x)|\bigr).$$

Then for each fixed x, $s \mapsto B_s(x)$ is entire. Moreover, for s in any compact set $K \subset \mathbb{C}$ we have uniform bounds

$$\sup_{s\in K} \|B_s\|_\infty \le M^{\sup_{s\in K} \Re(s)}, \qquad \sup_{s\in K} \|(B_s)'\|_\infty \le C_K < \infty, \tag{5}$$

since $(B_s)'(x) = s\,|f''(x)|^s\, \frac{f'''(x)}{f''(x)} \cdot \mathrm{sgn}(f''(x))$ and f'''/f'' is bounded by (3).

C.0.2. *A model endpoint-sampled series.*

Proof of Lemma 3. By Lemma 2, $Z_f(s) = 2^{-s} \sum_I |f''(\xi_I)|^s/(bd(b+d))^s$. Fix a small disk $D := \{s : |s - 2/3| \le \eta\}$ with $0 < \eta < 1/30$. Let $\sigma_- := \min_{s\in D} \Re(s) = 2/3 - \eta$.

For $s \in D$ and any $u, v \in [0,1]$, by the mean value theorem and (5),

$$\bigl||f''(u)|^s - |f''(v)|^s\bigr| \le \sup_{s\in D} \|(B_s)'\|_\infty\, |u - v| \le C_D\, |u - v|.$$

For a Farey interval $I = [c/d, a/b]$, we have $|\xi_I - a/b| \le |a/b - c/d| = 1/(bd)$. Hence for $s \in D$,

$$\left| \frac{|f''(\xi_I)|^s - |f''(a/b)|^s}{(bd(b+d))^s} \right| \le \frac{C_D}{(bd)\,(bd(b+d))^{\sigma_-}} = C_D\, b^{-(\sigma_-+1)} d^{-(\sigma_-+1)} (b+d)^{-\sigma_-}.$$

The right-hand side is independent of s and is summable over all $b, d \geq 1$ because at $\sigma_- = 2/3 - \eta$ the exponent on each of b, d is $\sigma_- + 1 > 1$. Therefore the series defining $Z_f(s) - Z_f^{\mathrm{end}}(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, the sum is holomorphic on the interior of D. □

C.1. **Auxiliary bounds for H_s and reduced-residue sampling.**

Lemma 13 (Pointwise bound for H_s)**.** *For $1/2 < \sigma := \Re(s) < 1$ and $u \in (0, 1]$ one has*

$$|H_s(u)| \leq C_s\, u^{-\sigma},$$

with C_s depending continuously on s on compact subsets of $\{1/2 < \Re(s) < 1\}$.

Proof. Write $H_s(u) = u^{-s}(1+u)^{-s} + \sum_{k\geq 1}(k+u)^{-s}(k+1+u)^{-s}$. The first term has modulus at most $u^{-\sigma}$. For the tail, $\sum_{k\geq 1}(k+u)^{-\sigma}(k+1+u)^{-\sigma} \leq \sum_{k\geq 1} k^{-2\sigma} < \infty$. Since $u^{-\sigma} \geq 1$ on $(0, 1]$, the tail is $\ll_s u^{-\sigma}$. □

Lemma 14 (Derivative bound for H_s)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Then there exists $C_K > 0$ such that for all $s \in K$ and all $u \in (0, 1]$,*

$$\left|\partial_u H_s(u)\right| \leq C_K\, u^{-\Re(s)-1}.$$

Proof. Differentiate termwise (absolute convergence holds uniformly for $u \in [\delta, 1]$, and the $k = 0$ term controls the singularity as $u \downarrow 0$):

$$\partial_u H_s(u) = -s \sum_{k\geq 0}(k+u)^{-s-1}(k+1+u)^{-s} - s \sum_{k\geq 0}(k+u)^{-s}(k+1+u)^{-s-1}.$$

For $u \in (0, 1]$ the $k = 0$ contributions satisfy $|(u)^{-s-1}(1+u)^{-s}| + |u^{-s}(1+u)^{-s-1}| \ll_K u^{-\Re(s)-1}$. For the tails $k \geq 1$ we bound each summand by $|s|\,(k+u)^{-\sigma-1}(k+1+u)^{-\sigma} + |s|\,(k+u)^{-\sigma}(k+1+u)^{-\sigma-1} \ll_K k^{-2\sigma-1}$, with $\sigma = \Re(s) > 1/2$, and $\sum_{k\geq 1} k^{-2\sigma-1} < \infty$ uniformly on K. Since $u^{-\sigma-1} \geq 1$ on $(0, 1]$, the tail is also $\ll_K u^{-\sigma-1}$. □

Lemma 15 (Reduced residues: H_s averages to its integral)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Then for $s \in K$ and $b \geq 1$,*

$$\sum_{\substack{1\leq r\leq b\\(r,b)=1}} H_s(r/b) = \varphi(b) \int_0^1 H_s(u)\, du + O_K\big(b^{\Re(s)}\tau(b)\big),$$

where $\tau(b)$ is the divisor function and the implied constant is uniform for $s \in K$.

Proof. By inclusion–exclusion,

$$\sum_{\substack{1\leq r\leq b\\(r,b)=1}} H_s(r/b) = \sum_{d|b} \mu(d) \sum_{1\leq k\leq b/d} H_s\left(\frac{k}{b/d}\right).$$

Write $N = b/d$ and denote $S_s(N) := \sum_{k=1}^{N} H_s(k/N)$. We prove

$$S_s(N) = N \int_0^1 H_s(u)\, du + O_K(N^{\Re(s)}), \tag{6}$$

uniformly for $s \in K$.

Decompose $H_s(u) = u^{-s}(1+u)^{-s} + R_s(u)$, where $R_s(u) := \sum_{k\geq 1}(k+u)^{-s}(k+1+u)^{-s}$.

Let $\sigma_- := \inf_{s\in K} \Re(s) > \frac{1}{2}$. Then for $u \in [0,1]$,

$$|(k+u)^{-s}(k+1+u)^{-s}| \leq k^{-2\Re(s)} \leq k^{-2\sigma_-},$$

and

$$\left|\partial_u\big((k+u)^{-s}(k+1+u)^{-s}\big)\right| \ll_K k^{-2\sigma_- -1}.$$

Since $\sum_{k\geq 1} k^{-2\sigma_-} < \infty$ and $\sum_{k\geq 1} k^{-2\sigma_- -1} < \infty$, both the series for R_s and for $\partial_u R_s$ converge uniformly on $[0,1]$, uniformly in $s \in K$. Hence $R_s \in C^1([0,1])$ and $\sup_{s\in K} \|R_s'\|_\infty < \infty$.

A standard Riemann-sum estimate for C^1 functions yields

$$\sum_{j=1}^{N} R_s(j/N) = N\int_0^1 R_s(u)\,du + O_K(1).$$

It remains to treat the singular part $G_s(u) := u^{-s}(1+u)^{-s}$. Write $G_s(u) = u^{-s} + u^{1-s}Q_s(u)$, where $Q_s(u) := \frac{(1+u)^{-s}-1}{u}$ extends continuously to $u = 0$ with $Q_s(0) = -s$. For $s \in K$, Q_s is C^1 on $[0,1]$ with norms bounded uniformly in s.

For

$$f_s(u) := u^{1-s}Q_s(u),$$

we have

$$f_s'(u) = (1-s)u^{-s}Q_s(u) + u^{1-s}Q_s'(u).$$

Since $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$, there exists $\sigma_+ < 1$ such that $\Re(s) \leq \sigma_+$ for all $s \in K$. As Q_s, Q_s' are uniformly bounded on $[0,1]$ for $s \in K$, it follows that

$$|f_s'(u)| \ll_K u^{-\sigma_+} + 1,$$

hence

$$\int_0^1 |f_s'(u)|\,du \ll_K 1.$$

Therefore, by the standard Riemann-sum estimate for absolutely continuous functions,

$$\sum_{j=1}^{N} f_s(j/N) = N\int_0^1 f_s(u)\,du + O_K(1).$$

So the contribution of $u^{1-s}Q_s(u)$ to (6) is $O_K(1)$.

Finally, for u^{-s} we use a direct Riemann-sum estimate (no zeta-function input). For $s \in K$ (hence $\Re(s) > \frac{1}{2}$ and $s \neq 1$), the function $x \mapsto x^{-s}$ is C^1 on $[1,\infty)$ and

$$\sum_{j=1}^{N} j^{-s} = \int_1^N x^{-s}\,dx + O_K(1).$$

Indeed,

$$\sum_{j=1}^{N} j^{-s} - \int_1^N x^{-s}dx = \sum_{j=1}^{N}\int_j^{j+1}\left(j^{-s} - x^{-s}\right)dx + O_K(1),$$

and the sum is bounded in modulus by $\sum_{j\geq 1}\int_j^{j+1}\int_j^{j+1}|(x^{-s})'|\,dx\,dt \ll_K \int_1^\infty x^{-\Re(s)-1}dx < \infty$. Multiplying by N^s yields

$$\sum_{j=1}^{N}(j/N)^{-s} = N^s\int_1^N x^{-s}dx + O_K(N^{\Re(s)}).$$

But $N^s\int_1^N x^{-s}dx = \frac{N-N^s}{1-s}$, so

$$\sum_{j=1}^{N}(j/N)^{-s} - N\int_0^1 u^{-s}du = -\frac{N^s}{1-s} + O_K(N^{\Re(s)}) = O_K(N^{\Re(s)}),$$

since $|N^s| = N^{\Re(s)}$. Combining the three pieces proves (6).

Returning to inclusion–exclusion,

$$\sum_{d|b}\mu(d)S_s(b/d) = \left(\sum_{d|b}\mu(d)\frac{b}{d}\right)\int_0^1 H_s + O_K\left(\sum_{d|b}(b/d)^{\Re(s)}\right).$$

The main coefficient equals $\varphi(b)$. The error satisfies $\sum_{d|b}(b/d)^{\Re(s)} \leq b^{\Re(s)}\tau(b)$. □

C.2. **Incomplete Kloosterman sums (Weil bound).** For integers n and $b \geq 1$ define the incomplete Kloosterman sum

$$K_b(n;R) := \sum_{\substack{1\leq r\leq R\\(r,b)=1}} e\left(\frac{n\overline{r}}{b}\right), \qquad e(x) := e^{2\pi i x}.$$

Lemma 16 (Completion bound for $K_b(n;R)$)**.** *For all integers n, all $b \geq 1$, and all $1 \leq R \leq b$,*

$$K_b(n;R) := \sum_{\substack{1\leq r\leq R\\(r,b)=1}} e\left(\frac{n\overline{r}}{b}\right) \ll (n,b)^{1/2}\,b^{1/2}\,\tau(b)^2\log(2b),$$

with an absolute implied constant.

In particular, if $(n,b)=1$, then for every $\varepsilon > 0$,

$$\max_{1\leq R\leq b}|K_b(n;R)| \ll_\varepsilon b^{1/2+\varepsilon}.$$

Proof. For $x \in \mathbb{Z}/b\mathbb{Z}$, define

$$w_R(x) := \sum_{m=1}^{R}\mathbf{1}_{x\equiv m \pmod b}.$$

Its discrete Fourier expansion is

$$w_R(r) = \frac{1}{b}\sum_{h \bmod b}\widehat{w}_R(h)\,e\left(\frac{hr}{b}\right), \qquad \widehat{w}_R(h) := \sum_{m=1}^{R}e\left(-\frac{hm}{b}\right).$$

Hence

$$K_b(n;R) = \sum_{\substack{r \bmod b\\(r,b)=1}} w_R(r)e\left(\frac{n\overline{r}}{b}\right) = \frac{1}{b}\sum_{h \bmod b}\widehat{w}_R(h)\,S(n,h;b),$$

where

$$S(n,h;b) := \sum_{\substack{r \bmod b \\ (r,b)=1}} e\left(\frac{n\overline{r}+hr}{b}\right)$$

is the classical Kloosterman sum.

We separate the zero frequency $h=0$. Since $\widehat{w}_R(0)=R$, we get

$$\frac{1}{b}\,\widehat{w}_R(0)\,S(n,0;b) = \frac{R}{b}\,c_b(n),$$

where $c_b(n)=S(n,0;b)$ is the Ramanujan sum.

Let $g=(n,b)$. Since

$$c_b(n) = \mu\left(\frac{b}{g}\right)\frac{\varphi(b)}{\varphi(b/g)},$$

we have

$$|c_b(n)| = \frac{\varphi(b)}{\varphi(b/g)} \le g = (n,b).$$

Hence

$$\left|\frac{R}{b}c_b(n)\right| \le (n,b),$$

and then it follows that

$$\left|\frac{1}{b}\,\widehat{w}_R(0)\,S(n,0;b)\right| \le (n,b).$$

Since $(n,b)\le (n,b)^{1/2}b^{1/2}$, this term is acceptable.

Now consider the nonzero frequencies $h\neq 0$. For representatives

$$-\frac{b}{2}\le h\le \frac{b}{2}, \qquad h\neq 0,$$

the geometric-series estimate gives

$$|\widehat{w}_R(h)| \ll \min\left(R, \frac{b}{|h|}\right).$$

Also, by the Weil bound for Kloosterman sums,

$$|S(n,h;b)| \ll \tau(b)\,(n,h,b)^{1/2}\,b^{1/2}.$$

Group the nonzero frequencies according to $d=(h,b)$. Then

$$(n,h,b)=(n,d).$$

Therefore

$$\sum_{\substack{h \bmod b \\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n,h;b)| \ll \tau(b)b^{1/2}\sum_{d|b}(n,d)^{1/2}\sum_{\substack{h \bmod b \\ h\neq 0 \\ (h,b)=d}} |\widehat{w}_R(h)|.$$

So it remains to bound the inner sum. Write $h=dh_1$. Then h_1 runs through a subset of nonzero residue classes modulo b/d, and hence

$$\sum_{\substack{h \bmod b \\ h\neq 0 \\ (h,b)=d}} |\widehat{w}_R(h)| \le \sum_{1\le |m|\le b/(2d)} \min\left(R, \frac{b}{d|m|}\right) \ll \frac{b}{d}\log(2b).$$

Thus

$$\sum_{\substack{h \bmod b \\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n,h;b)| \ll \tau(b)b^{1/2}\log(2b)\sum_{d|b}(n,d)^{1/2}\frac{b}{d}.$$

Dividing by b, we obtain

$$\frac{1}{b}\sum_{\substack{h \bmod b \\ h\neq 0}} |\widehat{w}_R(h)|\,|S(n,h;b)| \ll \tau(b)b^{1/2}\log(2b)\sum_{d|b}\frac{(n,d)^{1/2}}{d}.$$

Now

$$(n,d)^{1/2} \le (n,b)^{1/2},$$

so

$$\sum_{d|b}\frac{(n,d)^{1/2}}{d} \le (n,b)^{1/2}\sum_{d|b}\frac{1}{d} \le (n,b)^{1/2}\tau(b).$$

Hence the contribution of the nonzero frequencies is

$$\ll (n,b)^{1/2}b^{1/2}\tau(b)^2\log(2b).$$

Combining the zero and nonzero frequencies gives

$$K_b(n;R) \ll (n,b)^{1/2}b^{1/2}\tau(b)^2\log(2b),$$

as claimed.

Finally, if $(n,b)=1$, then

$$K_b(n;R) \ll b^{1/2}\tau(b)^2\log(2b) \ll_\varepsilon b^{1/2+\varepsilon},$$

since $\tau(b)^2\log(2b) \ll_\varepsilon b^\varepsilon$. □

C.3. **A Fejer approximation bound for C^1 functions.** Let F_N denote the Fejer kernel on the circle,

$$F_N(t) := \frac{1}{N}\left(\frac{\sin(\pi N t)}{\sin(\pi t)}\right)^2, \qquad t\in[-1/2,1/2],$$

extended 1-periodically.

Lemma 17 (First moment of the Fejer kernel)**.** *There exists an absolute constant $C>0$ such that for all $N\ge 2$,*

$$\int_{-1/2}^{1/2}|t|\,F_N(t)\,dt \le C\,\frac{\log N}{N}.$$

Consequently, if G is 1-periodic and C^1, then its Fejer mean $G*F_N$ satisfies

$$\|G-G*F_N\|_\infty \le C\,\frac{\log N}{N}\,\|G'\|_\infty.$$

Proof. Since $F_N\ge 0$ and $\int_{-1/2}^{1/2}F_N=1$,

$$G(x)-(G*F_N)(x) = \int_{-1/2}^{1/2}\big(G(x)-G(x-t)\big)F_N(t)\,dt.$$

By the mean value theorem, $|G(x)-G(x-t)|\le\|G'\|_\infty|t|$, giving the second inequality once the moment bound is proved.

For the moment bound, split the integral into $|t| \le 1/N$ and $1/N < |t| \le 1/2$. For $|t| \le 1/N$, use $F_N(t) \le N$ to get $\int_{|t|\le 1/N} |t| F_N(t) dt \ll N \int_0^{1/N} t\, dt \ll 1/N$. For $1/N < |t| \le 1/2$, use $|\sin(\pi t)| \gg |t|$ and $|\sin(\pi N t)| \le 1$ to obtain $F_N(t) \ll 1/(Nt^2)$. Hence $\int_{1/N<|t|\le 1/2} |t| F_N(t) dt \ll \frac{1}{N} \int_{1/N}^{1/2} \frac{dt}{t} \ll \frac{\log N}{N}$. □

C.4. **Proof of Proposition 2.**

Proof of Proposition 2. Fix $\varepsilon > 0$, and write $\sigma := \Re(s)$. Let

$$B_s(x) := |f''(x)|^s, \qquad A_r := H_s(r/b).$$

We may assume $b \ge 4$, since for bounded b the claim is absorbed into the implied constant. Set

$$N := R := \lfloor b^{1/2} \rfloor, \qquad \eta := \varepsilon/3.$$

Step 1: replace B_s by its Fejér mean. By (5), the family B_s is C^1 with norms bounded uniformly for $s \in K$. Hence Lemma 17 gives

$$\|B_s - (B_s)_N\|_\infty \ll_{f,K} \frac{\log(2N)}{N},$$

uniformly for $s \in K$. Therefore

$$\Sigma_b(s) = \sum_{\substack{1\le r\le b\\(r,b)=1}} A_r\, (B_s)_N(\overline{r}/b) + O_{f,K}\left(\frac{\log(2N)}{N} \sum_{\substack{1\le r\le b\\(r,b)=1}} |A_r| \right).$$

Now Lemma 13 yields

$$\sum_{\substack{1\le r\le b\\(r,b)=1}} |A_r| \le \sum_{r=1}^{b} |H_s(r/b)| \ll_K \sum_{r=1}^{b} (r/b)^{-\sigma} = b^\sigma \sum_{r=1}^{b} r^{-\sigma} \ll_K b,$$

uniformly for $s \in K$, since $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Thus

$$\Sigma_b(s) = \sum_{(r,b)=1} A_r\, (B_s)_N(\overline{r}/b) + O_{f,K}\big(b^{1/2} \log(2b)\big). \tag{7}$$

Step 2: expand the Fejér mean in Fourier modes. Since $(B_s)_N$ has the same zeroth Fourier coefficient as B_s,

$$(B_s)_N(x) = \widehat{B}_s(0) + \sum_{0<|n|<N} \widehat{(B_s)_N}(n) e(nx), \qquad \widehat{B}_s(0) = \int_0^1 B_s(v)\, dv.$$

Hence

$$\sum_{(r,b)=1} A_r\, (B_s)_N(\overline{r}/b) = \widehat{B}_s(0) \sum_{(r,b)=1} A_r + \sum_{0<|n|<N} \widehat{(B_s)_N}(n)\, S_b(n),$$

where

$$S_b(n) := \sum_{(r,b)=1} A_r\, e\left(\frac{n\overline{r}}{b}\right).$$

Step 3: the main term. By Lemma 15,

$$\sum_{(r,b)=1} A_r = \varphi(b) \int_0^1 H_s(u)\, du + O_K\big(b^\sigma \tau(b)\big),$$

uniformly for $s \in K$. Multiplying by

$$\widehat{B}_s(0) = \int_0^1 B_s(v)\, dv = \int_0^1 |f''(v)|^s\, dv$$

gives the claimed main term, plus an error

$$O_{f,K}\big(b^\sigma \tau(b)\big) \ll_{f,K,\eta} b^{\sigma+\eta} \ll_{f,K,\eta} b^{\frac{1+\sigma}{2}+\eta},$$

since $\sigma < 1$.

Step 4: bound the oscillatory sums $S_b(n)$. Fix n with $0 < |n| < N$. Split

$$S_b(n) = S_b^{\le R}(n) + S_b^{>R}(n),$$

where

$$S_b^{\le R}(n) := \sum_{\substack{1\le r\le R\\ (r,b)=1}} A_r\, e\!\left(\frac{n\overline{r}}{b}\right), \qquad S_b^{>R}(n) := \sum_{\substack{R<r\le b\\ (r,b)=1}} A_r\, e\!\left(\frac{n\overline{r}}{b}\right).$$

For the initial segment, Lemma 13 gives

$$|S_b^{\le R}(n)| \le \sum_{r\le R} |A_r| \ll_K b^\sigma \sum_{r\le R} r^{-\sigma} \ll_K b^\sigma R^{1-\sigma} \ll_K b^{\frac{1+\sigma}{2}}.$$

For the tail, define

$$a_r := \mathbf{1}_{(r,b)=1}\, e\!\left(\frac{n\overline{r}}{b}\right), \qquad T_n(t) := \sum_{1\le r\le t} a_r.$$

Then Lemma 16 gives

$$\max_{1\le t\le b} |T_n(t)| \ll_\eta (n,b)^{1/2} b^{1/2+\eta}.$$

By Abel summation,

$$S_b^{>R}(n) = H_s(1)\, T_n(b) - H_s(R/b)\, T_n(R) - \sum_{t=R}^{b-1} T_n(t)\Big(H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\Big).$$

Using Lemma 13,

$$|H_s(1)| \ll_K 1, \qquad |H_s(R/b)| \ll_K (R/b)^{-\sigma}.$$

Also, by the mean value theorem and Lemma 14,

$$\left|H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\right| \le \frac{1}{b} \sup_{u\in[t/b,(t+1)/b]} |\partial_u H_s(u)| \ll_K \frac{1}{b}\left(\frac{t}{b}\right)^{-\sigma-1} \ll_K b^\sigma t^{-\sigma-1}.$$

Hence

$$\sum_{t=R}^{b-1} \left|H_s\!\left(\frac{t+1}{b}\right) - H_s\!\left(\frac{t}{b}\right)\right| \ll_K b^\sigma \sum_{t\ge R} t^{-\sigma-1} \ll_K b^\sigma R^{-\sigma}.$$

Therefore

$$|S_b^{>R}(n)| \ll_{K,\eta} (n,b)^{1/2} b^{1/2+\eta}\Big(1 + (R/b)^{-\sigma} + b^\sigma R^{-\sigma}\Big).$$

Since $R \asymp b^{1/2}$, we have $(R/b)^{-\sigma} \asymp b^{\sigma/2}$ and $b^{\sigma}R^{-\sigma} \asymp b^{\sigma/2}$, so

$$|S_b^{>R}(n)| \ll_{K,\eta} (n,b)^{1/2} b^{\frac{1+\sigma}{2}+\eta}.$$

Combining this with the bound for $S_b^{\leq R}(n)$, we obtain

$$S_b(n) \ll_{K,\eta} (n,b)^{1/2}\, b^{\frac{1+\sigma}{2}+\eta} \qquad (0 < |n| < N). \tag{8}$$

Step 5: sum the Fourier modes. Since $B_s \in C^1$ uniformly for $s \in K$, we have

$$|\widehat{B}_s(n)| \ll_K \frac{\|(B_s)'\|_{L^1}}{|n|} \ll_{f,K} \frac{1}{|n|}, \qquad n \neq 0.$$

Moreover, for the Fejér mean,

$$|\widehat{(B_s)_N}(n)| \leq |\widehat{B}_s(n)|.$$

Thus, using (8),

$$\sum_{0<|n|<N} |\widehat{(B_s)_N}(n)\, S_b(n)| \ll_{f,K,\eta} b^{\frac{1+\sigma}{2}+\eta} \sum_{1\leq n<N} \frac{(n,b)^{1/2}}{n}.$$

To estimate the remaining sum, group integers by $d = (n,b)$:

$$\sum_{1\leq n<N} \frac{(n,b)^{1/2}}{n} \leq \sum_{d|b} \sum_{\substack{1\leq n<N \\ (n,b)=d}} \frac{d^{1/2}}{n}.$$

Writing $n = dm$, we get

$$\sum_{1\leq n<N} \frac{(n,b)^{1/2}}{n} \leq \sum_{d|b} d^{-1/2} \sum_{m<N/d} \frac{1}{m} \ll \log(2N) \sum_{d|b} d^{-1/2} \leq \tau(b)\log(2N).$$

Hence

$$\sum_{0<|n|<N} \widehat{(B_s)_N}(n)\, S_b(n) = O_{f,K,\eta}\Big(b^{\frac{1+\sigma}{2}+\eta}\tau(b)\log(2N)\Big) = O_{f,K,\varepsilon}\Big(b^{\frac{1+\sigma}{2}+\varepsilon}\Big),$$

since $\tau(b)\log(2N) \ll_\eta b^{2\eta}$.

Finally, the truncation error in (7) satisfies

$$b^{1/2}\log(2b) \ll_{K,\varepsilon} b^{\frac{1+\sigma}{2}+\varepsilon},$$

because $\sigma > \frac{1}{2}$. Putting everything together gives

$$\Sigma_b(s) = \varphi(b)\left(\int_0^1 H_s(u)\,du\right)\left(\int_0^1 |f''(v)|^s\,dv\right) + O_{f,K,\varepsilon}\Big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\Big),$$

uniformly for $s \in K$. □